Course: BCA

Python Libraries and Applications

Lab Sheet 2

Create a DataFrame using this code or you can access and Data Frame of your choice but including RollNo, Name, Department, Age and Marks features.

```
import pandas as pd

data = {
   "RollNo": [101, 102, 103, 104, 105, 106, 107, 108],
   "Name": [
      "Amit", "Neha", "Rahul", "Priya",
      "Rohan", "Kavya", "Arjun", "Simran"
   ],
   "Department": [
      "BCA", "BCA", "MCA", "BCA",
      "MCA", "MCA", "BCA", "MCA"
   ],
   "Age": [20, 21, 22, 20, 23, 22, 21, 24],
   "Marks": [85, 92, 67, 78, 88, 95, 55, 73]
}
df = pd.DataFrame(data)
```

Now solve the given queries:

Level 1 — Basic

Q1. Display the complete DataFrame.

Q2. Display the first 3 records.

Q3. Display the last 2 records.

Q4. Find the number of rows and columns.

Q5. Display all column names.

Q6. Display the data types of all columns.

Q7. Display only the `Name` column.

Q8. Display `Name` and `Marks`.

Level 2 — Selection and Filtering

Q9. Display students who scored more than 80.

Q10. Display students who scored less than 70.

Q11. Display students whose marks are between 70 and 90.

Q12. Display students belonging to the `BCA` department.

Q13. Display MCA students who scored more than 80.

Q14. Display students whose age is greater than 21.

Level 3 — Data Analysis

Q15. Calculate the average marks.

Q16. Find the maximum marks.

Q17. Find the minimum marks.

Q18. Find the median marks.

Q19. Find how many students scored above 75.

Q20. Find the average marks of BCA students.

Q21. Find the average marks of MCA students.

Q22. Find the highest-scoring student.

Level 4 — Data Manipulation

Q23. Add a new column called `Bonus` and give every student 5 marks.

Q24. Create a `FinalMarks` column:

```
FinalMarks = Marks + Bonus
```

Q25. Create a `Result` column:

```
Marks >= 40 → Pass
Marks < 40  → Fail
```

Q26. Create a `Grade` column using:

```
90 or above → A+
80–89       → A
70–79       → B
60–69       → C
Below 60    → D
```

Level 5 — Ranking

Q27. Sort students according to marks from highest to lowest.

Q28. Display the top 3 students.

Hint:

```
df.nlargest(3, "Marks")
```

Q29. Add a `Rank` column based on marks.

Hint:

```
df["Rank"] = df["Marks"].rank(
    ascending=False
)
```

Level 6 — Group Analysis

Q30. Find the number of students in each department.

Q31. Find the average marks of each department.

Q32. Find the maximum marks in each department.

Q33. Find the minimum marks in each department.

Try:

```
df.groupby("Department")["Marks"].agg(
    ["count", "mean", "max", "min"]
)
```

Challenge Question

Now think like a **data analyst**, not just a Python programmer.

Using the same DataFrame, produce a final merit list containing:

```
Rank
RollNo
Name
Department
Marks
Grade
Result
```

sorted by:

```
Marks → highest to lowest
```

Then export it as:

```
student_merit_list.xlsx
```

using:

```
df.to_excel(...)
```